power engineering
TRAINING SYSTEMS

SI Steam Tables for Reference in Power Engineering

For the use of **Students** and **Examination Candidates**, this document has been:

Approved by the Canadian Association of Chief Inspectors (ACI), as recommended by the Standardization of Power Engineer Examinations Committee (SOPEEC) and the Interprovincial Power Engineering Curriculum Committee (IPECC).

Symbols Used in Tables 1 and 2

h	specific enthalpy, kilojoules per kilogram
p	absolute pressure, kilopascals
s	specific entropy, kilojoules per kilogram degree kelvin
t	thermodynamic temperature, degrees Celsius
u	specific internal energy, kilojoules per kilogram
v	specific volume, cm^3 per gram (or x 10^{-3} m^3/kg or litres/kg)

Subscripts

f	refers to a property of liquid in equilibrium with vapour
g	refers to a property of vapour in equilibrium with liquid
fg	refers to a change by evaporation

Properties of Saturated Steam - Pressure Table

Table 1

		Specific Volume		Internal Energy			Enthalpy			Entropy		
Press	Temp	Sat.	Sat.	Sat.	Evap.	Sat.	Sat.	Evap.	Sat.	Sat.	Evap.	Sat.
k Pa	°C	Liquid	Vapor	Liquid		Vapor	Liquid		Vapor	Liquid		Vapor
p	t	V_f	V_g	U_f	U_{fg}	U_g	H_f	H_{fg}	H_g	S_f	S_{fg}	S_g
1.0	6.98	1.0002	129,208	29.30	2,355.7	2,385.0	29.30	2,484.9	2,514.2	0.1059	8.8697	8.9756
1.5	13.03	1.0007	87,980	57.71	2,338.6	2,393.3	57.71	2,470.6	2,525.3	0.1957	8.6322	8.8279
2.0	17.50	1.0013	67,004	73.48	2,326.0	2,399.5	73.48	2,460.0	2,533.5	0.2607	8.4629	8.7237
2.5	21.08	1.0020	54,254	88.48	2,315.9	2,404.4	88.49	2,451.6	2,540.0	0.3120	8.3311	8.6432
3.0	24.08	1.0027	45,665	101.04	2,307.5	2,408.5	101.05	2,444.5	2,545.5	0.3545	8.2231	8.5776
4.0	28.96	1.0040	34,800	121.45	2,293.7	2,415.2	121.46	2,432.9	2,554.4	0.4226	8.0520	8.4746
5.0	32.88	1.0053	28,192	137.81	2,282.7	2,420.5	137.82	2,423.7	2,561.5	0.4764	7.9187	8.3951
7.5	40.29	1.0079	19,238	168.78	2,261.7	2,430.5	168.79	2,406.0	2,574.8	0.5764	7.6750	8.2515
10	45.81	1.0102	14,674	191.82	2,246.1	2,437.9	191.83	2,392.8	2,584.7	0.6493	7.5009	8.1502
15	53.97	1.0141	10,022	225.92	2,222.8	2,448.7	225.94	2,373.1	2,599.1	0.7549	7.2536	8.0085
20	60.06	1.0172	7,649	251.38	2,205.4	2,456.7	251.40	2,358.3	2,609.7	0.8320	7.0766	7.9085
25	64.97	1.0199	6,204	271.90	2,191.2	2,463.1	271.93	2,346.3	2,618.2	0.8931	6.9383	7.8314
30	69.10	1.0223	5,229	289.20	2,179.2	2,468.4	289.23	2,336.1	2,625.3	0.9439	6.8247	7.7686
40	75.87	1.0265	3,993	317.53	2,159.5	2,477.0	317.58	2,319.2	2,636.8	1.0259	6.6441	7.6700
50	81.33	1.0300	3,240	340.44	2,143.4	2,483.9	340.49	2,305.4	2,645.9	1.0910	6.5029	7.5939
75	91.78	1.0373	2,217	384.31	2,112.4	2,496.7	384.39	2,278.6	2,663.0	1.2130	6.2434	7.4564
100	99.63	1.0432	1,694	417.36	2,088.7	2,506.1	417.46	2,258.0	2,675.5	1.3026	6.0568	7.3594
125	105.99	1.0483	1,374.9	444.19	2,069.3	2,513.5	444.32	2,241.0	2,685.4	1.3740	5.9104	7.2844
150	111.37	1.0528	1,159.3	466.94	2,052.7	2,519.7	467.11	2,226.5	2,693.6	1.4336	5.7897	7.2233
175	116.06	1.0568	1,003.6	486.80	2,038.1	2,524.9	486.99	2,213.6	2,700.6	1.4849	5.6868	7.1717
200	120.23	1.0605	885.7	504.49	2,025.0	2,529.5	504.70	2,201.9	2,706.7	1.5301	5.5970	7.1271
225	124.00	1.0640	793.3	520.47	2,013.1	2,533.6	520.72	2,191.3	2,712.1	1.5706	5.5173	7.0878
250	127.44	1.0672	718.7	535.10	2,002.1	2,537.2	535.37	2,181.5	2,716.9	1.6072	5.4455	7.0527
275	130.60	1.0703	657.3	548.59	1,991.9	2,540.5	548.89	2,172.4	2,721.3	1.6408	5.3801	7.0209
300	133.55	1.0732	605.8	561.15	1,982.4	2,543.6	561.47	2,163.8	2,725.3	1.6718	5.3201	6.9919
325	136.30	1.0759	562.0	572.90	1,973.5	2,546.4	573.25	2,155.8	2,729.0	1.7006	5.2646	6.9652
350	138.88	1.0786	524.3	583.95	1,965.0	2,548.9	584.33	2,148.1	2,732.4	1.7275	5.2130	6.9405
375	141.32	1.0811	491.4	594.40	1,956.9	2,551.3	594.81	2,140.8	2,735.6	1.7528	5.1647	6.9175
400	143.63	1.0836	462.5	604.31	1,949.3	2,553.6	604.74	2,133.8	2,738.6	1.7766	5.1193	6.8959
450	147.93	1.0882	414.0	622.77	1,934.9	2,557.6	623.25	2,120.7	2,743.9	1.8207	5.0359	6.8565
500	151.86	1.0926	374.9	639.68	1,921.6	2,561.2	640.23	2,108.5	2,748.7	1.8607	4.9606	6.8213
550	155.48	1.0967	342.7	655.32	1,909.2	2,564.5	655.93	2,097.0	2,753.0	1.8973	4.8920	6.7893
600	158.85	1.1006	315.7	669.90	1,897.5	2,567.4	670.56	2,086.3	2,756.8	1.9312	4.8288	6.7600
650	162.01	1.1044	292.7	683.56	1,886.5	2,570.1	684.28	2,076.0	2,760.3	1.9627	4.7703	6.7331
700	164.97	1.1080	272.9	696.44	1,876.1	2,572.5	697.22	2,066.3	2,763.5	1.9922	4.7158	6.7080
750	167.78	1.1115	255.6	708.64	1,866.1	2,574.7	709.47	2,057.0	2,766.4	2.0200	4.6647	6.6847
800	170.43	1.1148	240.4	720.22	1,856.6	2,576.8	721.11	2,048.0	2,769.1	2.0462	4.6166	6.6628
850	172.96	1.1181	227.0	731.27	1,847.4	2,578.7	732.22	2,039.4	2,771.6	2.0710	4.5711	6.6421
900	175.38	1.1212	215.0	741.83	1,868.6	2,580.5	742.83	2,031.1	2,773.9	2.0946	4.5280	6.6226
950	177.69	1.1243	204.2	751.95	1,830.2	2,582.1	753.02	2,023.1	2,776.1	2.1172	4.4869	6.6041
1,000	179.91	1.1273	194.44	761.68	1,822.0	2,583.6	762.81	2,015.3	2,778.1	2.1387	4.4478	6.5865
1,100	184.09	1.1330	177.53	780.09	1,806.3	2,586.4	781.34	2,000.4	2,781.7	2.1792	4.3744	6.5536
1,200	187.99	1.1385	163.33	797.29	1,791.5	2,588.8	798.65	1,986.2	2,784.8	2.2166	4.3067	6.5233
1,300	191.64	1.1438	151.25	813.44	1,777.5	2,591.0	814.93	1,972.7	2,787.6	2.2515	4.2438	6.4953
1,400	195.07	1.1489	140.84	828.70	1,764.1	2,592.8	830.30	1,959.7	2,790.0	2.2842	4.1850	6.4693
1,500	198.32	1.1539	131.77	843.16	1,751.3	2,594.5	844.89	1,947.3	2,792.2	2.3150	4.1298	6.4448
1,750	205.76	1.1656	113.49	876.46	1,721.4	2,597.8	878.50	1,917.9	2,796.4	2.3851	4.0044	6.3896

Note: 100k Pa = 1 Bar = 100,000N/m^2

Properties of Saturated Steam - Pressure Table

Table 1 (continued...)

		Specific Volume		Internal Energy			Enthalpy			Entropy		
Press	Temp	Sat.	Sat.	Sat.	Evap.	Sat.	Sat.	Evap.	Sat.	Sat.	Evap.	Sat.
k Pa	°C	Liquid	Vapor	Liquid		Vapor	Liquid		Vapor	Liquid		Vapor
p	t	V_f	V_g	U_f	U_{fg}	U_g	H_f	H_{fg}	H_g	S_f	S_{fg}	S_g
2,000	212.42	1.1767	99.63	906.44	1,693.8	2,600.3	908.79	1,890.7	2,799.5	2.4474	3.8935	6.3409
2,250	218.45	1.1872	88.75	933.83	1,668.2	2,602.0	936.49	1,865.2	2,801.7	2.5035	3.7937	6.2972
2,500	223.99	1.1973	79.98	959.11	1,644.0	2,603.1	962.11	1,841.0	2,803.1	2.5547	3.7028	6.2575
3,000	233.90	1.2165	66.68	1,004.78	1,599.3	2,604.1	1,008.42	1,795.7	2,804.2	2.6457	3.5412	6.1869
3,500	242.60	1.2347	57.07	1,045.43	1,558.3	2,603.7	1,049.75	1,753.7	2,803.4	2.7253	3.4000	6.1253
4,000	250.40	1.2522	49.78	1,082.31	1,520.0	2,602.3	1,087.31	1,714.1	2,801.4	2.7964	3.2737	6.0701
5,000	263.99	1.2859	39.44	1,147.81	1,449.3	2,597.1	1,154.23	1,640.1	2,794.3	2.9202	3.0532	5.9734
6,000	275.64	1.3187	32.44	1,205.44	1,384.3	2,589.7	1,213.35	1,571.0	2,784.3	3.0267	2.8625	5.8892
7,000	285.88	1.3513	27.37	1,257.55	1,323.0	2,580.5	1,267.00	1,505.1	2,772.1	3.1211	2.6922	5.8133
8,000	295.06	1.3842	23.52	1,305.57	1,264.2	2,569.8	1,316.64	1,441.3	2,758.0	3.2068	2.5364	5.7432
9,000	303.40	1.4178	20.48	1,350.51	1,207.3	2,557.8	1,363.26	1,378.9	2,742.1	3.2858	2.3915	5.6772
10,000	311.06	1.4524	18.026	1,393.04	1,151.4	2,544.4	1,407.56	1,317.1	2,724.7	3.3596	2.2544	5.6141
11,000	318.15	1.4886	15.987	1,433.7	1,096.0	2,529.8	1,450.1	1,255.5	2,705.6	3.4295	2.1233	5.5527
12,000	324.75	1.5267	14.263	1,473.0	1,040.7	2,513.7	1,491.3	1,193.6	2,684.9	3.4962	1.9962	5.4924
13,000	330.93	1.5671	12.780	1,511.1	985.0	2,496.1	1,531.5	1,130.7	2,662.2	3.5606	1.8718	5.4323
14,000	336.75	1.6107	11.485	1,548.6	928.2	2,476.8	1,571.1	1,066.5	2,637.6	3.6232	1.7485	5.3717
15,000	342.24	1.6581	10.337	1,585.6	869.8	2,455.5	1,610.5	1,000.0	2,610.5	3.6848	1.6249	5.3098
16,000	347.44	1.7107	9.306	1,622.7	809.0	2,431.7	1,650.1	930.6	2,580.6	3.7461	1.4994	5.2455
17,000	352.37	1.7702	8.364	1,660.2	744.8	2,405.0	1,690.3	856.9	2,547.2	3.8079	1.3698	5.1777
18,000	357.06	1.8397	7.489	1,698.9	675.4	2,374.3	1,732.0	777.1	2,509.1	3.8715	1.2329	5.1044
19,000	361.54	1.9243	6.657	1,739.9	598.1	2,338.1	1,776.5	688.0	2,464.5	3.9388	1.0839	5.0228
20,000	365.81	2.036	5.834	1,785.6	507.5	2,293.0	1,826.3	583.4	2,409.7	4.039	0.9130	4.9269
21,000	369.89	2.207	4.952	1,842.1	388.5	2,230.6	1,888.4	446.2	2,334.6	4.1075	0.6938	4.8013
22,000	373.80	2.742	3.568	1,961.9	125.2	2,087.1	2,022.2	143.4	2,165.6	4.3110	0.2216	4.5327
22,090	374.14	3.155	3.155	2,029.6	0	2,029.6	2,099.3	0	2,099.3	4.4298	0	4.4298

Note: 100k Pa = 1 Bar = 100,000N/m^2

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Properties of Saturated Steam - Temperature Table

Table 2

		Specific Volume		Internal Energy			Enthalpy			Entropy		
Temp °C	Press. k Pa	Sat. Liquid	Sat. Vapor	Sat. Liquid	Evap.	Sat. Vapor	Sat. Liquid	Evap.	Sat. Vapor	Sat. Liquid	Evap.	Sat. Vapor
t	p	V_f	V_g	U_f	U_{fg}	U_g	H_f	H_{fg}	H_g	S_f	S_{fg}	S_g
0	0.6109	1.0002	206,278	0.000	2,375.4	2,375.3	0.000	2,501.4	2,501.3	0.000	9.1566	9.1565
5	0.8721	1.0001	147,120	20.97	2,361.3	2,382.3	20.98	2,489.6	2,510.6	0.0761	8.9496	9.0257
10	1.2276	1.0004	106,379	42.00	2,347.2	2,389.2	42.01	2,477.7	2,519.8	0.1510	8.7498	8.9008
15	1.7051	1.0009	77,926	62.99	2,333.1	2,396.1	62.99	2,465.9	2,528.9	0.2245	8.5569	8.7814
20	2.339	1.0018	57,791	83.95	2,319.0	2,402.9	83.96	2,454.1	2,538.1	0.2966	8.3706	8.6672
25	3.169	1.0029	43,360	104.88	2,304.9	2,409.8	104.89	2,442.3	2,547.2	0.3674	8.1905	8.5580
30	4.246	1.0043	32,894	125.78	2,290.8	2,416.6	125.79	2,430.5	2,556.3	0.4369	8.0164	8.4533
35	5.628	1.0060	25,216	146.67	2,276.7	2,423.4	146.68	2,418.6	2,565.3	0.5053	7.8478	8.3531
40	7.384	1.0078	19,523	167.56	2,262.6	2,430.1	167.57	2,406.7	2,574.3	0.5725	7.6845	8.2570
45	9.593	1.0099	15,258	188.44	2,248.4	2,436.8	188.45	2,394.8	2,583.2	0.6387	7.5261	8.1648
50	12.349	1.0121	12,032	209.32	2,234.2	2,443.5	209.33	2,382.7	2,592.1	0.7038	7.3725	8.0763
60	19.940	1.0172	7,671	251.11	2,205.5	2,456.6	251.13	2,358.5	2,609.6	0.8312	7.0784	7.9096
70	31.19	1.0228	5,042	292.95	2,176.6	2,469.6	292.98	2,333.8	2,626.8	0.9549	6.8004	7.7553
80	47.39	1.0291	3,407	334.86	2,147.4	2,482.2	334.91	2,308.8	2,643.7	1.0753	6.5369	7.6122
90	70.14	1.0360	2,361	376.85	2,117.7	2,494.5	376.92	2,283.2	2,660.1	1.1925	6.2866	7.4791
100	101.35	1.0435	1,672.9	418.94	2,087.6	2,506.5	419.04	2,257.0	2,676.1	1.3069	6.0480	7.3549
110	143.27	1.0516	1,210.2	461.14	2,057.0	2,518.1	461.30	2,230.2	2,691.5	1.4185	5.8202	7.2387
120	198.53	1.0603	891.9	503.50	2,025.8	2,529.3	503.71	2,202.6	2,706.3	1.5276	5.6020	7.1296
130	270.1	1.0697	668.5	546.02	1,993.9	2,539.9	546.31	2,174.2	2,720.5	1.6344	5.3925	7.0269
140	361.3	1.0797	508.9	588.74	1,961.3	2,550.0	589.13	2,144.7	2,733.9	1.7391	5.1908	6.9299
150	475.8	1.0905	392.8	631.68	1,927.9	2,559.5	632.20	2,114.3	2,746.5	1.8418	4.9960	6.8379
160	617.8	1.1020	307.1	674.87	1,893.5	2,568.4	675.55	2,082.6	2,758.1	1.9427	4.8075	6.7502
170	791.7	1.1143	242.8	718.33	1,858.1	2,576.5	719.21	2,049.5	2,768.7	2.0419	4.6244	6.6662
180	1002.1	1.1274	194.05	762.09	1,821.6	2,583.7	763.22	2,015.0	2,778.2	2.1396	4.4461	6.5857
190	1254.4	1.1414	156.54	806.19	1,783.8	2,590.0	807.62	1,978.8	2,786.4	2.2359	4.2720	6.5079
200	1553.8	1.1565	127.36	850.65	1,744.7	2,595.3	852.45	1,940.7	2,793.2	2.3309	4.1014	6.4323
210	1906.2	1.1726	104.41	895.53	1,703.9	2,599.5	897.76	1,900.7	2,798.5	2.4248	3.9337	6.3585
220	2318	1.1900	86.19	940.87	1,661.5	2,602.4	943.62	1,858.5	2,802.1	2.5178	3.7683	6.2861
230	2795	1.2088	71.58	986.74	1,617.2	2,603.9	990.12	1,813.8	2,804.0	2.6099	3.6047	6.2146
240	3344	1.2291	59.76	1,033.21	1,570.8	2,604.0	1,037.32	1,766.5	2,803.8	2.7015	3.4422	6.1437
250	3973	1.2512	50.13	1,080.39	1,522.0	2,602.4	1,085.36	1,716.2	2,801.5	2.7927	3.2802	6.0730
260	4688	1.2755	42.21	1,128.39	1,470.6	2,599.0	1,134.37	1,662.5	2,796.9	2.8838	3.1181	6.0019
270	5499	1.3023	35.64	1,177.36	1,416.3	2,593.7	1,184.51	1,605.2	2,789.7	2.9751	2.9551	5.9301
280	6412	1.3321	30.17	1,227.46	1,358.7	2,586.1	1,235.99	1,543.6	2,779.6	3.0668	2.7903	5.8571
290	7436	1.3656	25.57	1,278.92	1,297.1	2,576.0	1,289.07	1,477.1	2,766.2	3.1594	2.6227	5.7821
300	8581	1.4036	21.67	1,332.0	1,231.0	2,563.0	1,344.0	1,404.9	2,749.0	3.2534	2.4511	5.7045
310	9856	1.4474	18.350	1,387.1	1,159.4	2,546.4	1,401.3	1,326.0	2,727.3	3.3493	2.2737	5.6230
320	11274	1.4988	15.488	1,444.6	1,080.9	2,525.5	1,461.5	1,238.6	2,700.1	3.4480	2.0882	5.5362
330	12845	1.5607	12.996	1,505.3	993.7	2,498.9	1,525.3	1,140.6	2,665.9	3.5507	1.8909	5.4417
340	14586	1.6379	10.797	1,570.3	894.3	2,464.6	1,594.2	1,027.9	2,622.0	3.6594	1.6763	5.3357
350	16513	1.7403	8.813	1,641.9	776.6	2,418.4	1,670.6	893.4	2,563.9	3.7777	1.4335	5.2112
360	18651	1.8925	6.945	1,725.2	626.3	2,351.5	1,760.5	720.5	2,481.0	3.9147	1.1379	5.0526
370	21030	2.213	4.925	1,844.0	384.5	2,228.5	1,890.5	441.6	2,332.1	4.1106	0.6865	4.7971
374.136	22090	3.155	3.155	2,029.6	0	2,029.6	2,099.3	0	2,099.3	4.4298	0	4.4298

Note: 100k Pa = 1 Bar = 100,000N/m^2

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Table of Properties for Superheated Steam

Table 3

Abs Press. kPa (Sat. Temp)		Temperature Degrees Celsius											
		100	**150**	**200**	**250**	**300**	**400**	**500**	**600**	**700**	**800**	**900**	**1,000**
	v	17,196	19,512	21,825	24,136	26,445	31,063	35,679	40,295	44,911	49,526	54,141	58,757
10	*h*	2,687.5	2,783.0	2,879.5	2,977.3	3,076.5	3,279.6	3,489.1	3,705.4	3,928.7	4159.0	4,396.4	4,640.6
(45.81)	*s*	8.4479	8.6882	8.9038	9.1002	9.2813	9.6077	9.8978	10.1608	10.4028	10.6281	10.8396	11.0393
	v	5,715	6,493	7,267	8,040	8,811	10,351	1,1891	13,430	14,969	16,508	18,047	19,585
30	*h*	2,685.0	2,781.5	2,878.6	2,976.7	3,076.0	3,279.2	3,488.9	3,705.3	3,928.6	4,158.9	4,396.4	4,640.5
(69.10)	*s*	7.9357	8.1785	8.3952	8.5923	8.7736	9.1003	9.3906	9.6537	9.8957	10.1210	10.3325	10.5322
	v	3,418	3,889	4,356	4,820	5,284	6,209	7,134	8,057	8,981	9,904	10,828	11,751
50	*h*	2,682.5	2,780.1	2,877.7	2,976.0	3,075.5	3,278.9	3,488.7	3,705.1	3,928.5	4,158.9	4,396.3	4,640.5
(81.33)	*s*	7.6947	7.9401	8.1580	8.3556	8.5373	8.8642	9.1546	9.4178	9.6599	9.8852	10.0967	10.2964
	v	1,695.8	1,936.4	2,172	2,406	2,639	3,103	3,565	4,028	4,490	4,952	5,414	5,875
100	*h*	2,676.2	2,776.4	2,875.3	2,974.3	3,074.3	3,278.2	3,488.1	3,704.7	3,928.2	4,158.6	4,396.1	4,640.3
(99.63)	*s*	7.3614	7.6134	7.8343	8.0333	8.2158	8.5435	8.8342	9.0976	9.3398	9.5652	9.7767	9.9764
	v		959.6	1,080.3	1,198.8	1,316.2	1,549.3	1,781.4	2,013	2,244	2,475	2,706	2,937
200	*h*		2,768.8	2,870.5	2,971.0	3,071.8	3,276.6	3,487.1	3,704.0	3,927.6	4,158.2	4,395.8	4,640.0
(120.23)	*s*		7.2795	7.5066	7.7086	7.8926	8.2218	8.5133	8.7770	9.0194	9.2449	9.4566	9.6563
	v		633.9	716.3	796.4	875.3	1,031.5	1,186.7	1,341.4	1,495.7	1,649.9	1,804.1	1,958.1
300	*h*		2761.0	2,865.6	2,967.6	3,069.3	3,275.0	3,486.0	3,703.2	3,927.1	4,157.8	4,395.4	4,639.7
(133.55)	*s*		7.0778	7.3115	7.5166	7.7022	8.0330	8.3251	8.5892	8.8319	9.0576	9.2692	9.4690
	v			424.9	474.4	522.6	617.3	710.9	804.1	896.9	989.6	1,082.2	1,174.7
500	*h*			2,855.4	2,960.7	3,064.2	3,271.9	3,483.9	3,701.7	3,925.9	4,156.9	4,394.7	4,639.1
(151.86)	*s*			7.0592	7.2709	7.4599	7.7938	8.0873	8.3522	8.5952	8.8211	9.0329	9.2328
	v			206.0	232.7	257.9	306.6	354.1	401.1	447.8	494.3	540.7	587.1
1,000	*h*			2,827.9	2,942.6	3,051.2	3,263.9	3,478.5	3,697.9	3,923.1	4,154.7	4,392.9	4,637.6
(179.91)	*s*			6.6940	6.9247	7.1229	7.4651	7.7622	8.0290	8.2731	8.4966	8.7118	8.9119
	v				111.44	125.47	151.20	175.68	199.60	223.2	246.7	270.0	293.3
2,000	*h*				2,902.5	3,023.5	3,247.6	3,467.6	3,690.1	3,917.4	4,150.3	4,389.4	4,634.6
(212.43)	*s*				6.5453	6.7664	7.1271	7.4317	7.7024	7.9487	8.1765	8.3895	8.5901
	v				70.58	81.14	99.36	116.19	132.43	148.38	164.14	197.80	195.41
3,000	*h*				2,855.8	2,993.5	3,230.9	3,456.5	3,682.3	3,911.7	4,145.9	4,385.9	4,631.6
(233.90)	*s*				6.2872	6.5390	6.9212	7.2338	7.5085	7.7571	7.9862	8.1999	8.4009
	v					58.84	73.41	86.43	98.95	110.95	122.87	134.69	146.45
4,000	*h*					2,960.7	3,213.6	3,445.3	3,674.4	3,905.9	4,141.5	4,382.3	4,628.7
(250.40)	*s*					6.3615	6.7690	7.0901	7.3688	7.6198	7.8502	8.0647	8.2662
	v					45.32	57.81	68.57	78.69	88.49	98.11	107.62	117.07
5,000	*h*					2,924.5	3,195.7	3,433.8	3,666.5	3,900.1	4,137.1	4,378.8	4,625.7
(263.99)	*s*					6.2084	6.6459	6.9759	7.2589	7.5122	7.7440	7.9593	8.1612
	v					36.16	47.39	56.65	65.25	73.52	81.60	89.58	97.49
6,000	*h*					2,884.2	3,177.2	3,422.2	3,658.4	3,894.2	4,132.7	4,375.3	4,622.7
(275.64)	*s*					6.0674	6.5408	6.8803	7.1677	7.4234	7.6566	7.8727	8.0751

Note: 100k Pa = 1 Bar = 100,000N/m^2

Table of Properties for Superheated Steam

Table 3 (continued...)

Abs Press. kPa (Sat. Temp)		Temperature Degrees Celsius											
		100	150	200	250	300	400	500	600	700	800	900	1,000
	v					29.47	39.93	48.14	55.65	62.83	69.81	76.69	83.50
7,000	*h*					2,838.4	3,158.1	3,410.3	3,650.3	3,888.3	4,128.2	4,371.8	4,619.8
(285.88)	*s*					5.9305	6.4478	6.7975	7.0894	7.3476	7.5822	7.7991	8.0020
	v					24.26	34.32	41.75	48.45	54.81	60.97	67.02	73.01
8,000	*h*					2,785.0	3,138.3	3,398.3	3,642.0	3,882.4	4,123.8	4,368.3	4,616.9
(295.06)	*s*					5.7906	6.3634	6.7240	7.0206	7.2812	7.5173	7.7351	7.9384
	v						29.93	36.77	42.85	48.57	54.09	59.50	64.85
9,000	*h*						3,117.8	3,386.1	3,633.7	3,876.5	4,119.3	4,364.8	4,614.0
(303.40)	*s*						6.2854	6.6576	6.9589	7.2221	7.4596	7.6783	7.8821
	v						26.41	32.79	38.37	43.58	48.59	53.49	58.32
10,000	*h*						3,096.5	3,373.7	3,625.3	3,870.5	4,114.8	4,361.2	4,611.0
(311.06)	*s*						6.2120	6.5966	6.9029	7.1687	7.4077	7.6262	7.8315
	v						23.51	29.52	34.70	39.50	44.09	48.57	52.98
11,000	*h*						3,074.3	3,361.0	3,616.9	3,864.5	4,110.4	4,357.7	4,608.1
(318.15)	*s*						6.1420	6.5400	6.8514	7.1199	7.3605	7.5808	7.7856
	v						21.08	26.80	31.64	36.10	40.34	44.47	48.53
12,000	*h*						3,051.3	3,348.2	3,608.3	3,858.4	4,105.9	4,354.2	4,605.3
(324.75)	*s*						6.0747	6.4871	6.8037	7.0749	7.3170	7.5382	4.7435
	v						19.007	24.50	29.05	33.22	37.17	41.00	44.77
13,000	*h*						3,027.2	3,335.2	3,599.7	3,852.3	4,101.4	4,350.7	4,602.4
(330.93)	*s*						6.0091	6.4371	6.7591	7.0331	7.2767	7.4988	7.7047
	v						17.216	22.52	26.83	30.75	34.45	38.03	41.54
14,000	*h*						3,001.9	3,322.0	3,591.1	3,846.2	4,096.9	4,347.2	4,599.5
(336.75)	*s*						5.9448	6.3897	6.7172	6.9939	7.2392	7.4622	7.6685
	v						15.649	20.80	24.91	28.61	32.10	35.46	38.75
15,000	*h*						2,975.5	3,308.6	3,582.3	3,840.1	4,092.4	4,343.8	4,596.6
(342.24)	*s*						5.8811	6.3443	6.6776	6.9572	7.2040	7.4279	7.6348
	v						14.262	19.296	23.23	26.74	30.03	33.20	36.30
16,000	*h*						2,947.6	3,294.9	3,573.5	3,833.9	4,087.8	4,340.3	4,593.8
(347.44)	*s*						5.8175	6.3007	6.6399	6.9224	7.1708	7.3957	7.6031
	v						13.021	17.967	21.74	25.09	28.22	31.22	34.14
17,000	*h*						2,918.2	3,281.1	3,564.6	3,827.7	4,083.3	4,336.8	4,590.9
(352.37)	*s*						5.7536	6.2587	6.6040	6.8894	7.1395	7.3653	7.5732
	v						11.901	16.784	20.42	23.62	26.60	29.45	32.23
18,000	*h*						2,887.0	3,267.0	3,555.6	3,821.5	4,078.8	4,333.3	4,588.1
(357.06)	*s*						5.6887	6.2181	6.5696	6.8580	7.1098	7.3365	7.5779
	v						10.881	15.724	19.241	22.31	25.15	27.87	30.51
19,000	*h*						2,853.8	3,252.7	3,546.6	3,815.3	4,074.3	4,329.8	4,585.3
(361.54)	*s*						5.6224	6.1786	6.5366	6.8281	7.0814	7.3091	7.5181

Note: 100k Pa = 1 Bar = 100,000N/m^2

Table of Properties for Superheated Steam

Table 3 (continued...)

Abs Press. kPa (Sat. Temp)		Temperature Degrees Celsius											
		100	150	200	250	300	400	500	600	700	800	900	1,000
	v						9.942	14.768	18.178	21.13	23.85	26.45	28.97
20,000	*h*						2,818.1	3,238.2	3,537.6	3,809.0	4,069.7	4,326.4	4,582.5
(365.81)	*s*						5.5540	6.1401	6.5048	6.7993	7.0544	7.2830	7.4925
	v						9.071	13.903	17.216	20.06	22.68	25.16	27.57
21,000	*h*						2,779.6	3,223.5	3,528.4	3,802.8	4,065.2	4,322.9	4,579.6
(369.89)	*s*						5.4826	6.1026	6.4741	6.7717	7.0285	7.2581	7.4681
	v						8.253	13.115	16.341	19.092	21.61	23.99	26.30
22,000	*h*						2,737.6	3,208.6	3,519.2	3,796.5	4,060.6	4,319.5	4,576.8
(373.80)	*s*						5.4074	6.0658	6.4444	6.7451	7.0036	7.2342	7.4448
	v						7.478	12.394	15.543	18.206	20.63	22.92	25.14
23,000	*h*						2,691.2	3,193.4	3,510.0	3,790.2	4,056.2	4,316.0	4,574.0
	s						5.3269	6.0297	6.4155	6.7195	6.9798	7.2113	7.4224
	v						6.732	11.733	14.811	17.393	19.735	21.944	24.08
24,000	*h*						2,639.4	3,178.0	3,500.7	3,783.8	4,051.6	4,312.5	4,571.3
	s						5.2393	5.9942	6.3875	6.6947	6.9567	7.1892	7.4009
	v						6.004	11.123	14.137	16.646	18.912	21.045	23.10
25,000	*h*						2,580.2	3,162.4	3,491.4	3,777.5	4,047.1	4,309.1	4,568.5
	s						5.1418	5.9592	6.3602	6.6707	6.9345	7.1680	7.3802
	v						5.280	10.560	13.516	15.957	18.153	20.214	22.20
26,000	*h*						2,510.9	3,146.6	3,482.0	3,777.1	4,042.5	4,305.7	4,565.7
	s						5.0306	5.9248	6.3336	6.6475	6.9130	7.1475	7.3602
	v						4.550	10.038	12.941	15.319	17.450	19.446	21.37
27,000	*h*						2,428.2	3,130.5	3,472.5	3,764.8	4,038.0	4,302.2	4,562.9
	s						4.9003	5.8907	6.3077	6.6248	6.8922	7.1276	7.3409
	v						3.831	9.553	12.407	14.727	16.797	18.732	20.59
28,000	*h*						2,330.7	3,114.3	3,463.0	3,758.4	4,033.4	4,298.8	4,560.2
	s						4.7494	5.8570	6.2823	6.6029	6.8720	7.1084	7.3223
	v						3.214	9.101	11.910	14.175	16.190	18.068	19.869
29,000	*h*						2,231.1	3,097.8	3,453.5	3,752.0	4,028.8	4,295.4	4,557.4
	s						4.5962	5.8236	6.2574	6.5814	6.8523	7.0898	7.3042
	v						2.790	8.678	11.446	13.661	15.623	17.448	19.196
30,000	*h*						2,151.1	3,081.1	3,443.9	3,745.6	4,024.2	4,291.9	4,554.7
	s						4.4728	5.7905	6.2331	6.5606	6.8332	7.0718	7.2867
	v						2.527	8.283	11.012	13.180	15.093	16.868	18.567
31,000	*h*						2,095.0	3,064.1	3,434.3	3,739.2	4,019.7	4,288.5	4,551.9
	s						4.3856	5.7576	6.2092	6.5402	6.8147	7.0542	7.2697
	v						2.361	7.912	10.606	12.729	14.597	16.325	17.977
32,000	*h*						2,055.9	3,047.0	3,424.6	3,732.8	4,015.1	4,285.1	4,549.2
	s						4.3239	5.7250	6.1858	6.5203	6.7966	7.0372	7.2533

Note: 100k Pa = 1 Bar = 100,000N/m^2

Table of Properties for Superheated Steam

Table 3 (continued...)

Abs Press. kPa (Sat. Temp)		Temperature Degrees Celsius											
		100	150	200	250	300	400	500	600	700	800	900	1,000
	v						2.247	7.564	10.224	12.306	14.130	15.815	17.423
33,000	*h*						2,027.3	3,029.6	3,415.0	3,726.4	4,010.6	4,281.7	4,546.5
	s						4.2781	5.6925	6.1628	6.5008	6.7790	7.0206	7.2372
	v						2.164	7.236	9.865	11.908	13.691	15.335	16.901
34,000	*h*						2,005.3	3,012.1	3,405.2	3,719.9	4,006.0	4,278.3	4,543.8
	s						4.2421	5.6603	6.1401	6.4818	6.7618	7.0044	7.2216
	v						2.100	6.927	9.527	11.533	13.278	14.833	16.410
35,000	*h*						1,987.6	2,994.4	3,395.5	3,713.5	4001.5	4,274.9	4,541.1
	s						4.2126	5.6282	6.1179	6.4631	6.7450	6.9886	7.7.2064
	v						2.048	6.636	9.208	11.179	12.887	14.456	15.946
36,000	*h*						1,972.8	2976.4	3,385.7	3,707.0	4,996.9	4,271.5	4,538.4
	s						4.1875	5.5962	6.0960	6.4448	6.7285	6.9732	7.1916
	v						2.005	6.361	8.907	10.844	12.518	14.052	15.507
37,000	*h*						1,960.2	2958.4	3,375.9	3,700.6	3,992.4	4,268.1	4,535.7
	s						4.1658	5.5644	6.0744	6.4269	6.7125	6.9582	7.1771
	v						1.9679	6.101	8.621	10.527	12.169	13.670	15.092
38,000	*h*						1,949.2	2,940.1	3,366.1	3,694.1	3,987.8	4,264.7	4,533.0
	s						4.1466	5.5328	6.0531	6.4093	6.6967	6.9435	7.1630

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